

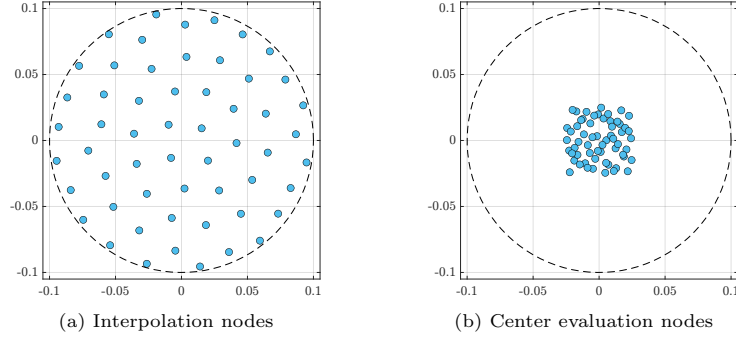


Figure 1: Distribution of interpolation and evaluation nodes within a circle of radius 0.1.

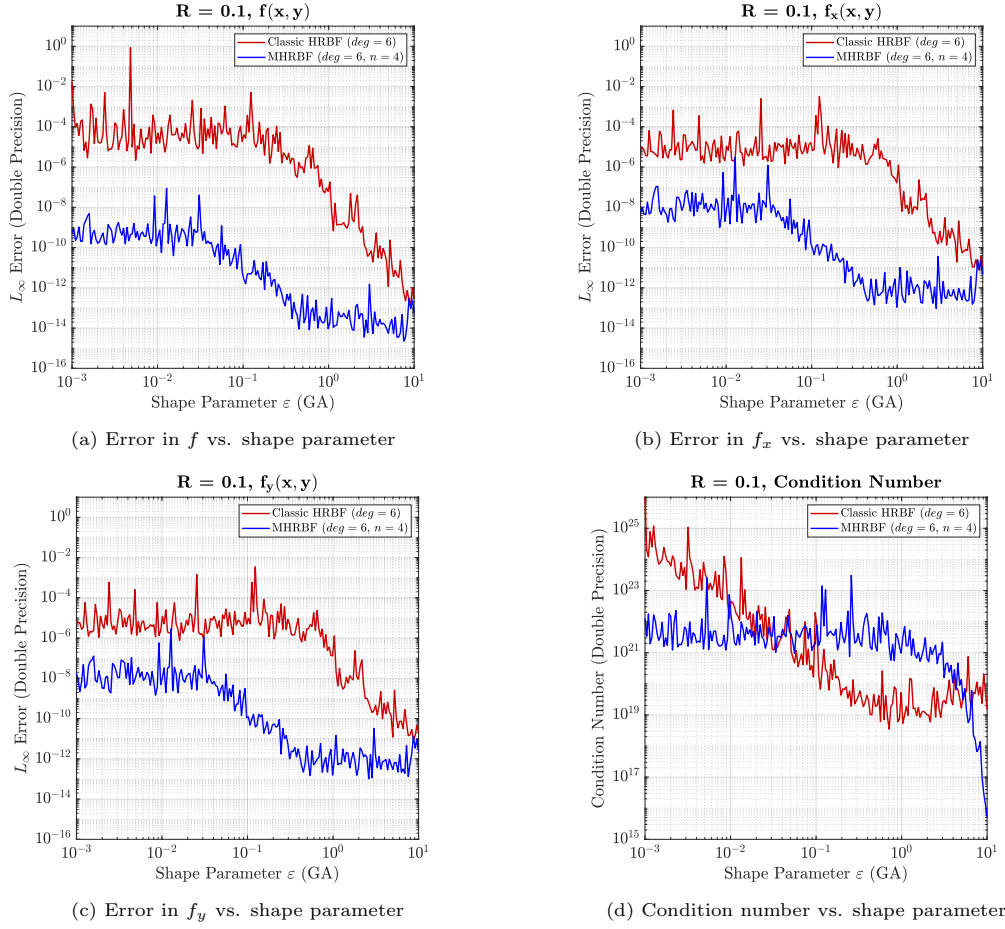


Figure 2: Comparison of Accuracy and Numerical Stability for HRBF and MHRBF: The errors (function, f , and its first derivatives, f_x and f_y) and condition number computed in double precision for GA kernel, as functions of shape parameter ε , with polynomial degree of 6 and $n = 4$ for MHRBF.

The results in Figure 2 reveal significant accuracy improvements with MHRBF compared to HRBF. Since the error trends for ∇f closely follow that of f , the subsequent analysis focuses primarily on the function interpolation. Figure 2a demonstrates that the MHRBF achieves low error over a large range of ε values, with the error $\sim 10^{-14}$ for all $\varepsilon \gtrsim 0.5$, despite the fact that the condition number of the MHRBF being